

	Work Instruction	Doc. Revision	01
	AXI V810 STD X-Ray Tube Alignment Procedure	Effective Date	2 Oct 2016

This procedure is applicable to below models:

V810 STD	☒	S1	☒	S2
V810 XXL	☐	S1	☐	S2
V810 S2EX	☐			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	13 May 2015	First Release	CHIN CHEE YAN
01	2 Oct 2016	Template update & Content update	ONG KEAN SHENG

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- Go to path "C:\Program Files\AXI System\v810\<Rev>\<config>" and open "software.config" file.
Search key "saveallconfirmationimage", set to "True" and save.

<Rev> represent to GUI version.

#<config> only applicable for V810 standard system (S1/S2)

#<xxlconfig> applicable for V810 XXL system (S1/S2)

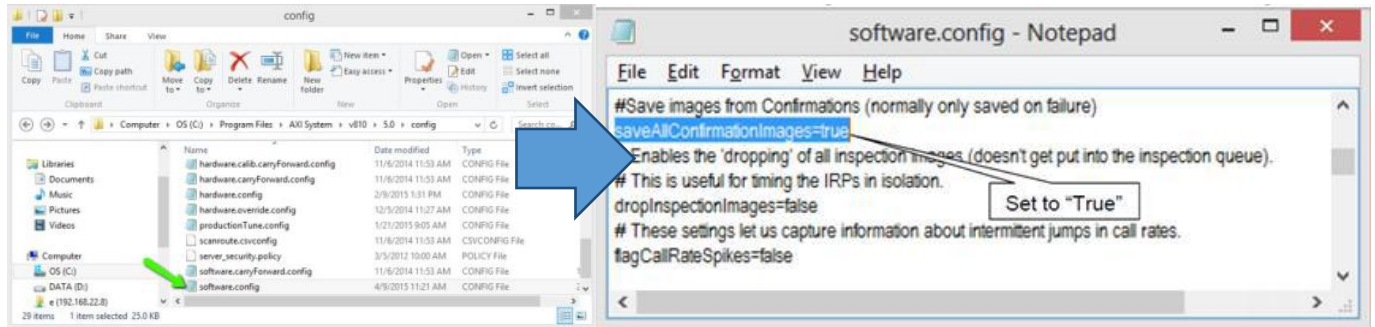


Figure 1 Software Config

- Go to Connected Processor, click Stop RMS, Upgrade RMS and Restart RMS as Figure 2.

Services → Subsystem Status & Control → Connected Processor → Stop RMS → Upgrade RMS

→ Restart RMS

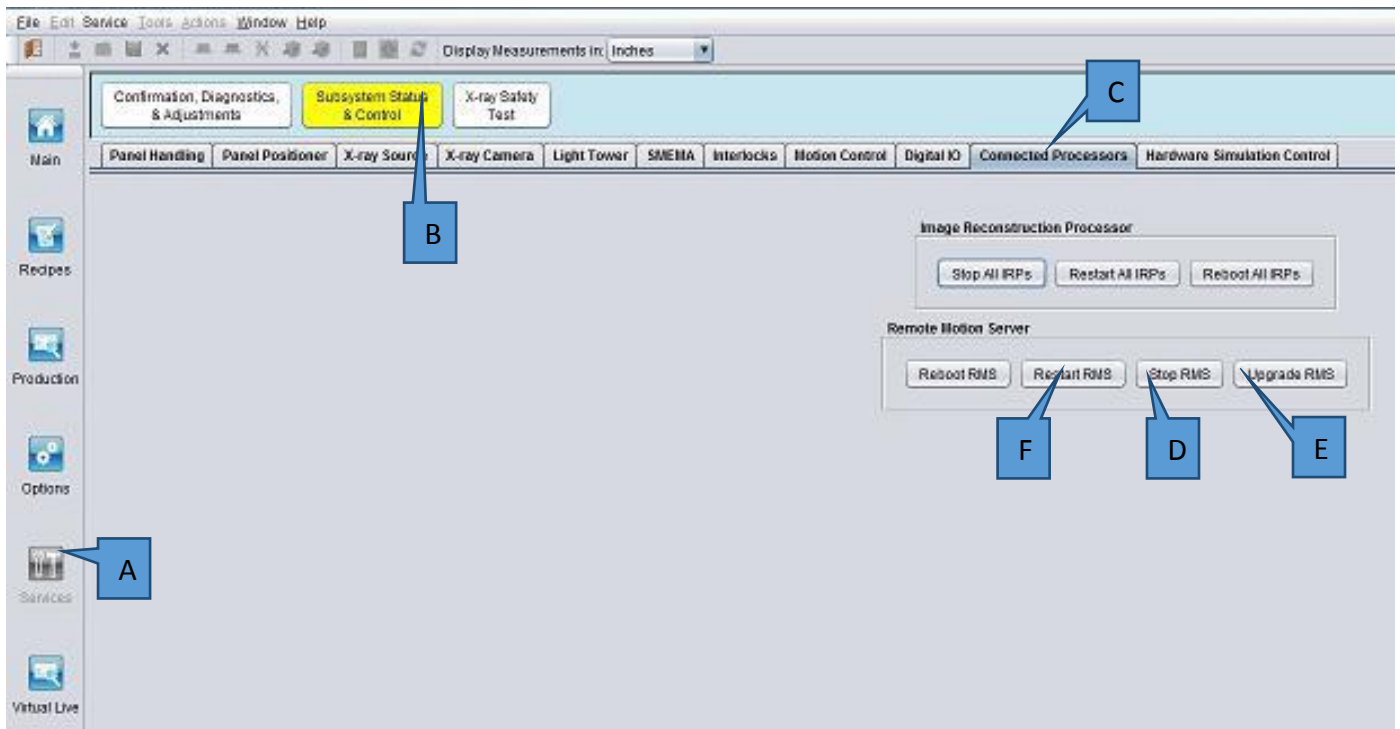


Figure 2 Connector Processor

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3. Quit and restart V810 software. Go to Digital I/O tab and initialize as show in Figure 3.

Services → Subsystem Status & Control → Digital I/O → Initialize

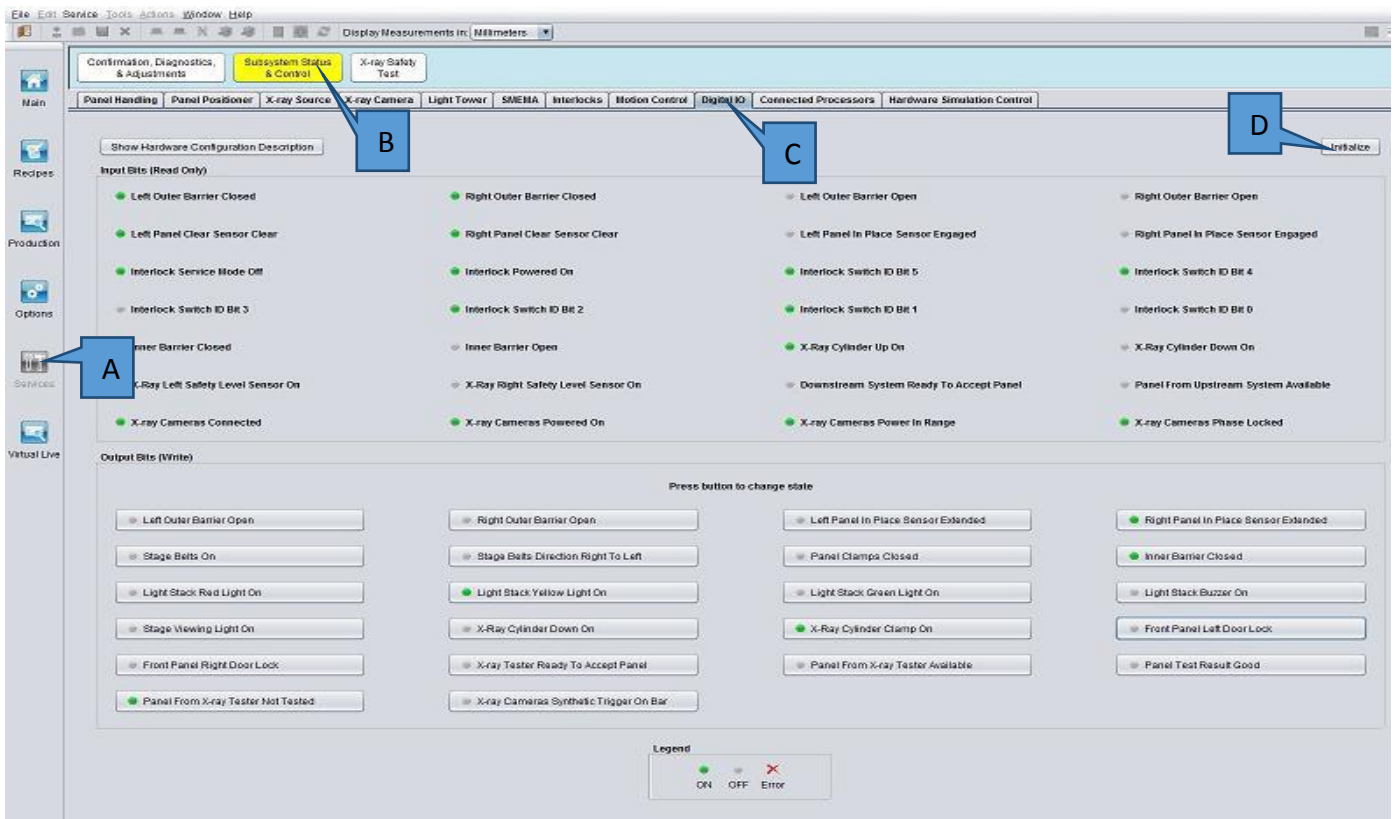


Figure 3 Digital IO

4. Go to Panel Handling tab and initialize as show in Figure 4. Inner Barrier in Open position,

Services → Subsystem Status & Control → Panel Handling → Inner Barrier Open Position

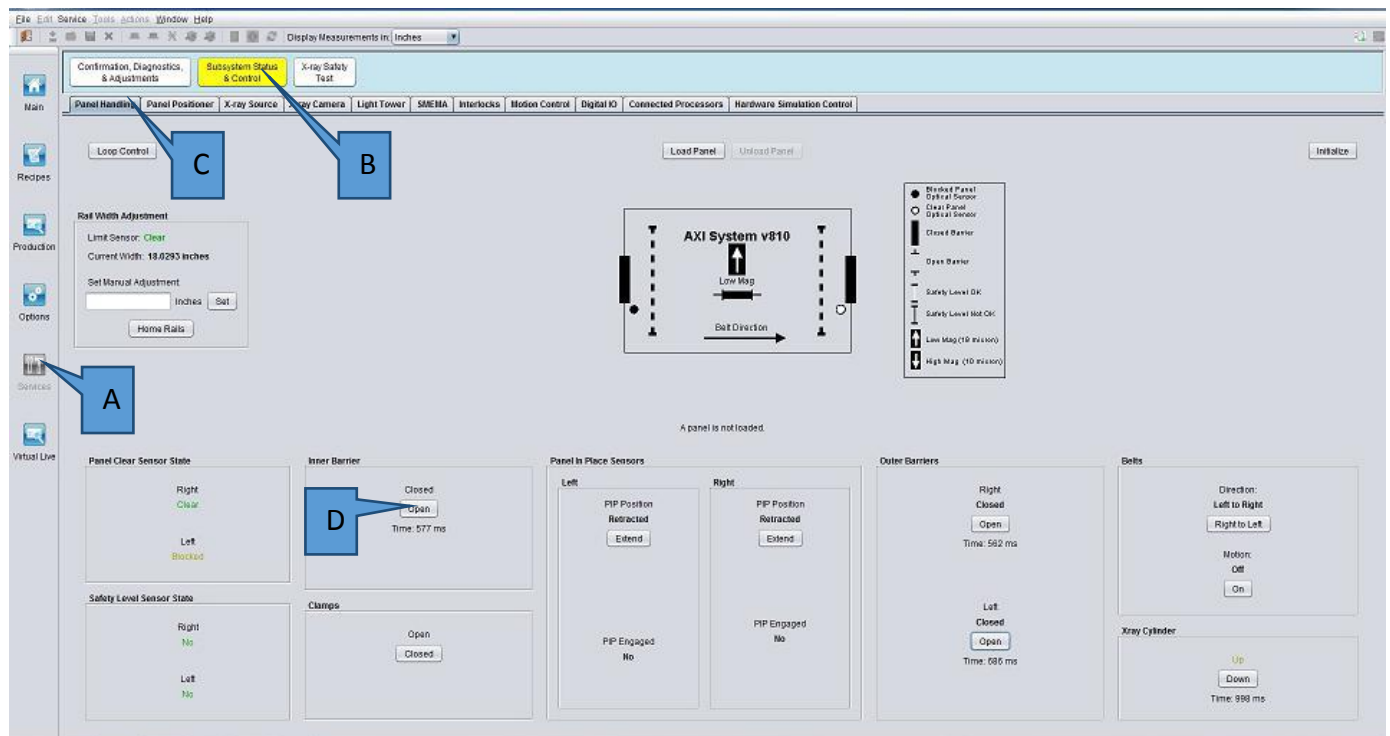


Figure 4 Panel Handling

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- Go to Confirmation, Diagnostic and Adjustment tab and perform 1 loop full CD&A task to obtain latest log file as show in Figure 5.

Services → Confirmation, Diagnostic and Adjustment → Scheduled Tasks → Run 1 loop

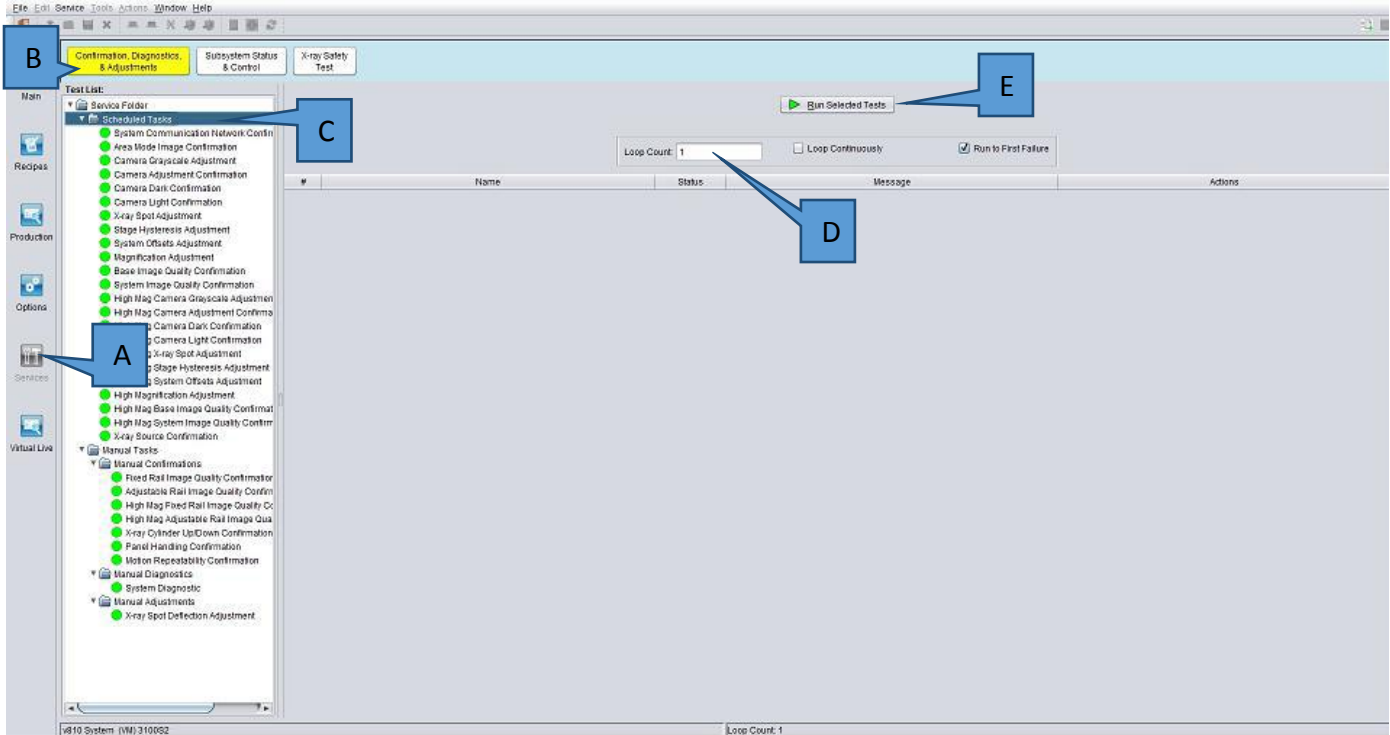


Figure 5 CD & A

- Once complete, go to path “C:\Program Files\AXI System\v810\5.0\log\calib\XraySpot” and check for the updated (latest modify date) cam_0.png (X-Ray spot image file). Backup the files as origin reference.

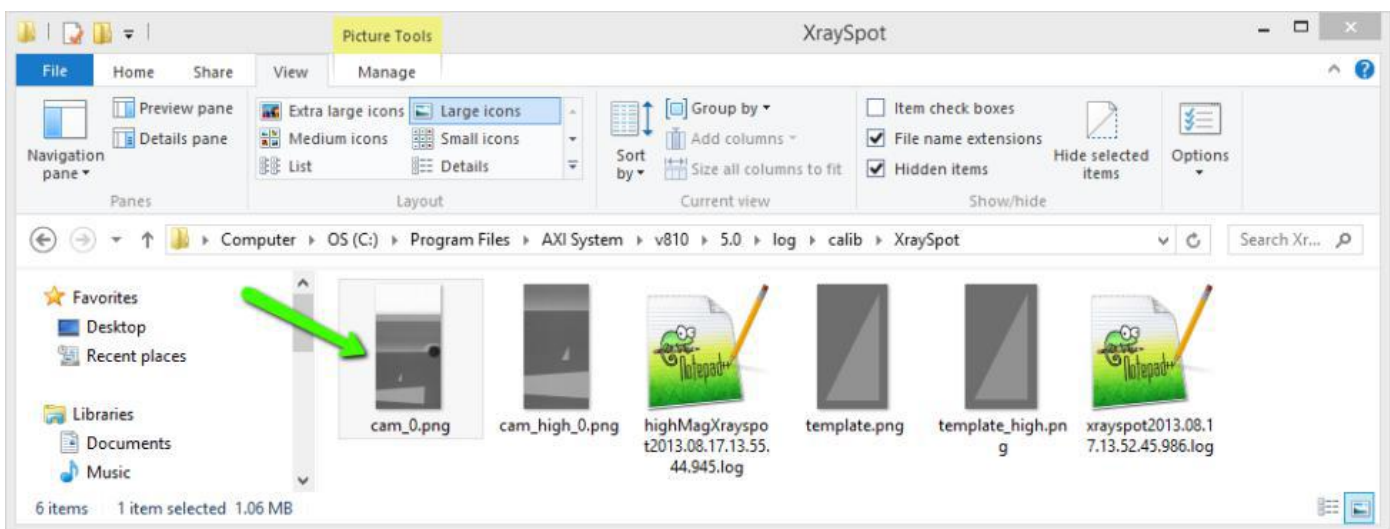


Figure 6 Xray Spot

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7. May refer to below golden sample of X ray spot & adjust current tube position as similar as golden sample.

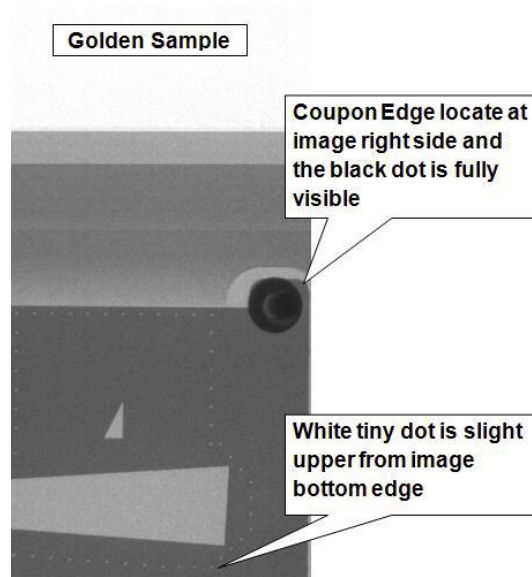


Figure 7 X ray spot [GOLDEN SAMPLE]

8. **Turn off X-ray source**, access from top access panel and unscrew (Tamper proof screw) as shown. Gently open the access door.

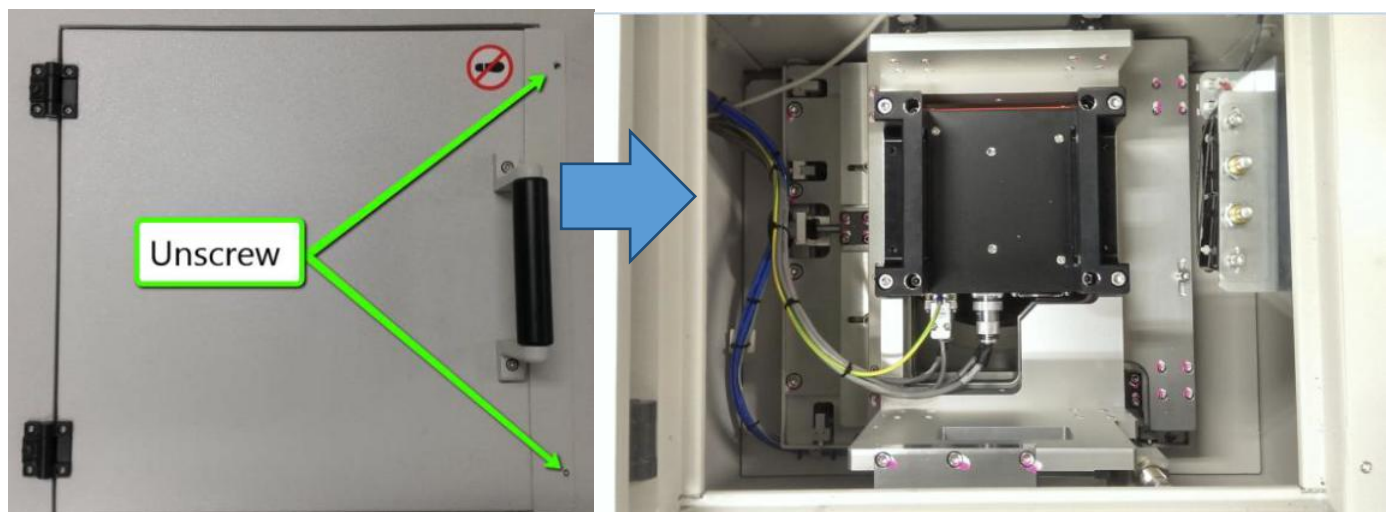


Figure 8 Top access door

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9. May refer to below figure for part identification.

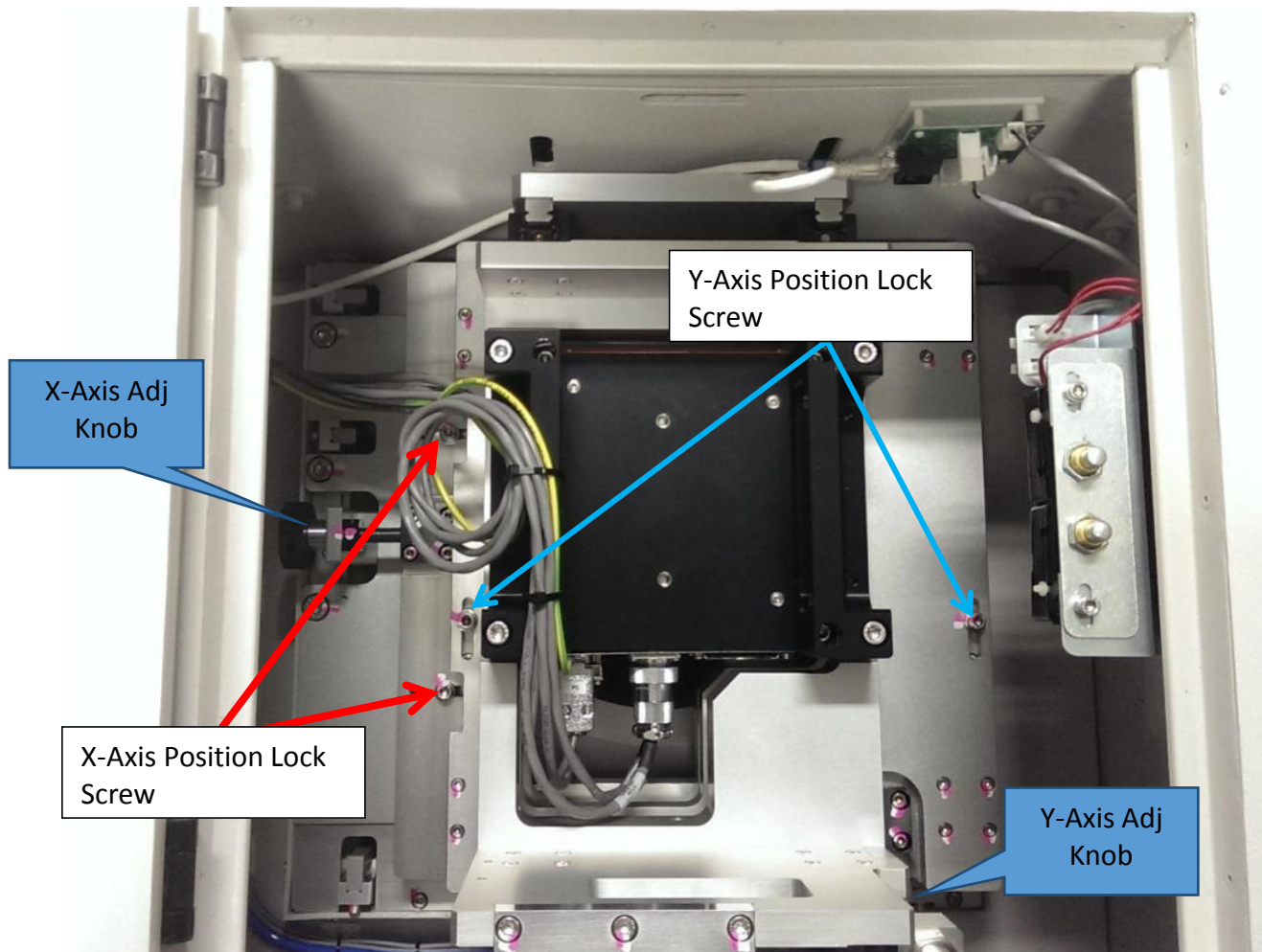


Figure 9 X ray Tube assembly

10. In order to move the tube position, those lock screws need to be loosen. For the old tube wall design, need to loosen the screws and nuts also to move the tube position.

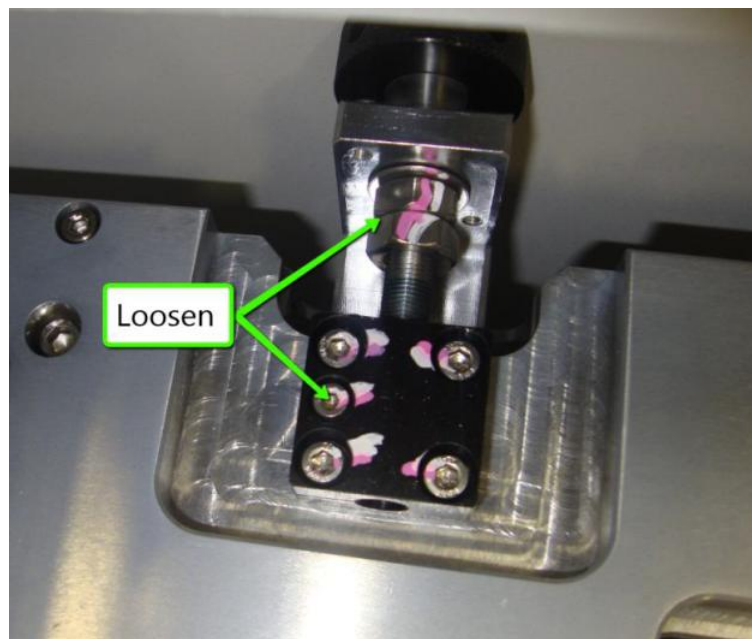


Figure 10 Tube Adjustment Knob

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11. Compare the existing X-ray spot image with the golden sample and move the tube according.

Term,

- If image need to move to the left, physical tube position need move to machine right site.
- If image need to move to the top, physical tube position need move to machine front site.
- If image need to move to the right, physical tube position need move to machine left site.
- If image need to move to the bottom, physical tube position need move to machine rear site.

Sample Case Study

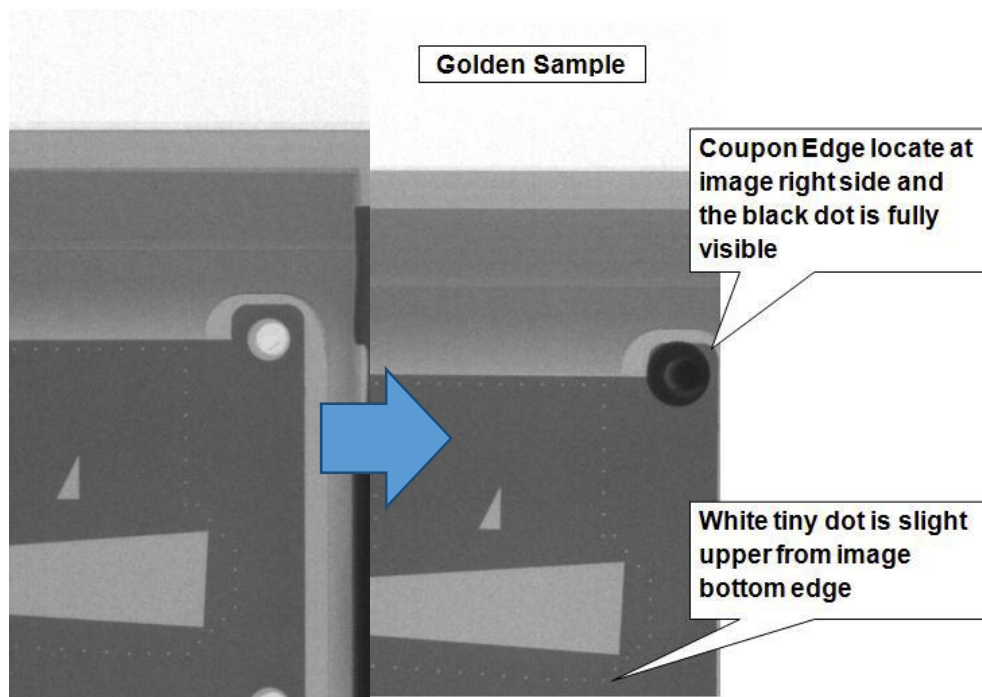


Figure 11 X Ray Spot

In this case, tube X-Axis need move to machine left side and Y-axis move to machine front side to get the optimize X-Ray spot same as golden sample. Turn the Adj knob to move (# 1 turn of knob = 40 mils (1 mm)) the tube position. Check on the screw track to verify the actual movement direction (X and Y axis)

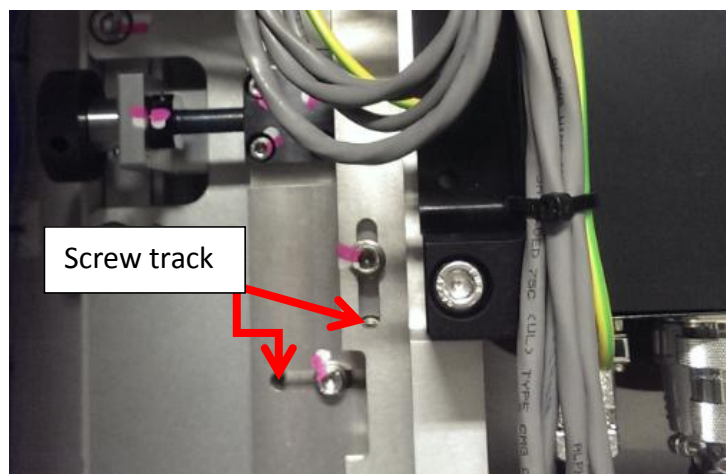


Figure 12 Locking mechanism

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12. Turn the knob 1 turn each time, close the top access door, ON the X-ray source and back to CD&A tab. Select X-Ray Spot Adjustment task and click on “Run test without running Dependent tests”.

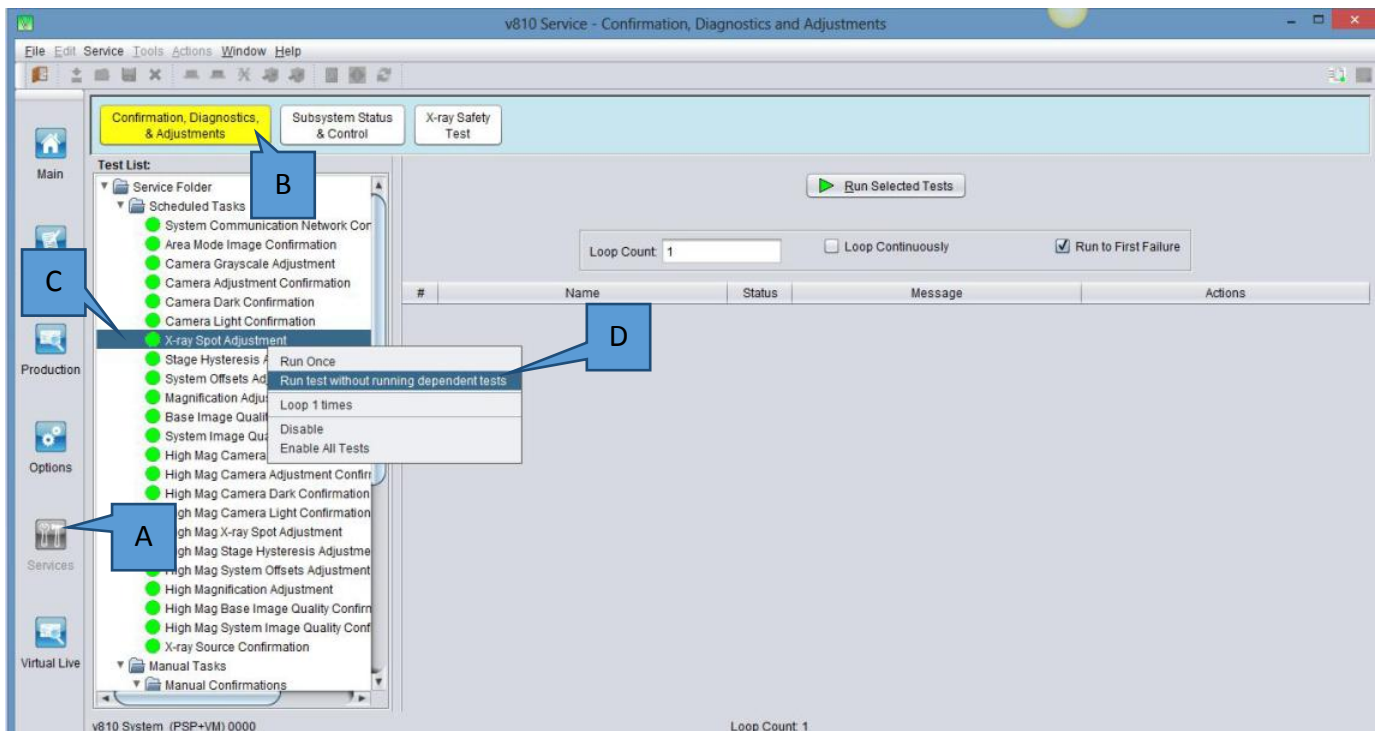


Figure 13 CD & A

13. Once task complete, check the latest X-ray post image. Repeat step 6 ~ 12 until actual X-Ray spot as similar as golden sample.

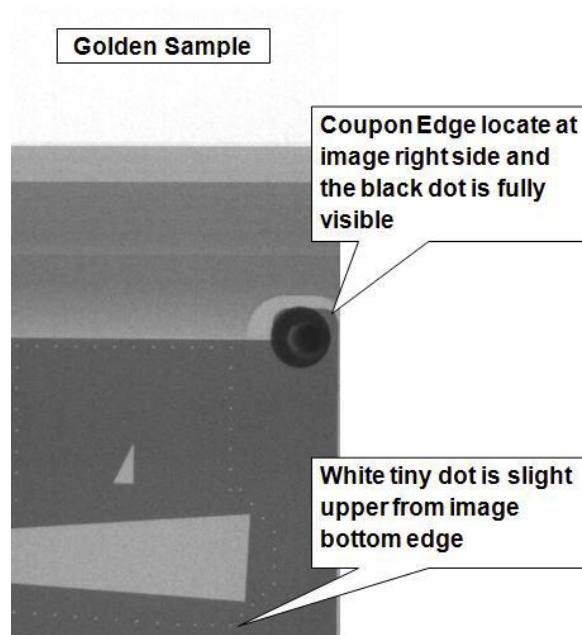


Figure 14 X Ray Spot [GOLDEN SAMPLE]

14. Once done, tighten the locking screws and nuts and close the top access door with screw. Go to Confirmation, Diagnostic and Adjustment tab and perform 5 loop full CD&A task with select schedule task.